

SIGNIFICANCE / HYPOTHESIS TESTING

ARISES IN REAL-WORLD SITUATIONS

LIKE: • IS PROCESS A BETTER THAN
PROCESS B?

- DID STATISTICS OF A PROCESS CHANGE?
- DOES PROCESS HAVE TARGET DISTRIBUTION X?

(DETECTION AND ESTIMATION COURSE
GOES INTO THESE MORE DEEPLY)

SIGNIFICANCE TESTING: WE ONLY HAVE
THE NULL HYPOTHESIS H_0 .

WE WANT TO ACCEPT IT AS TRUE
OR REJECT IT AS FALSE WITH
SOME SIGNIFICANCE LEVEL.

⊗ SIMILAR TO
CONFIDENCE
INTERVAL.

TYPE I ERROR: WE REJECTED H_0 WHEN
IT WAS ACTUALLY TRUE

} SIGNIFICANCE
TESTING

TYPE II ERROR: WE ACCEPTED H_0 WHEN
IT WAS ACTUALLY FALSE.

↑ SIGNIFICANCE TESTING CAN'T
REASON ABOUT THIS

α = TYPE I ERROR

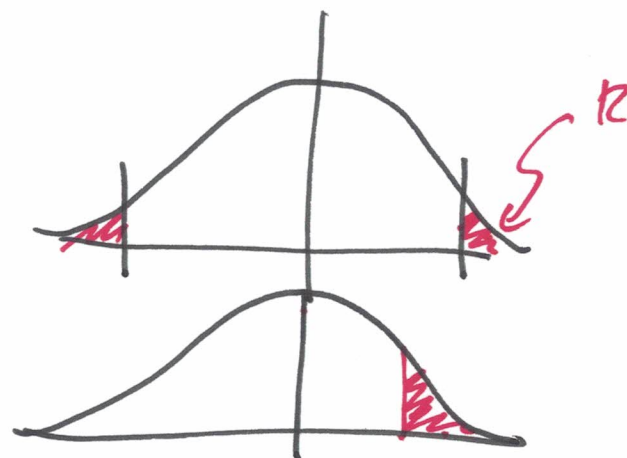
R : A REGION SUCH THAT

MEASUREMENT $\in R \Rightarrow$ REJECT H_0 AT
THE $\alpha\%$ SIGNIFICANCE
LEVEL

MEASUREMENT $\notin R \Rightarrow$ ACCEPT H_0 AT THE...

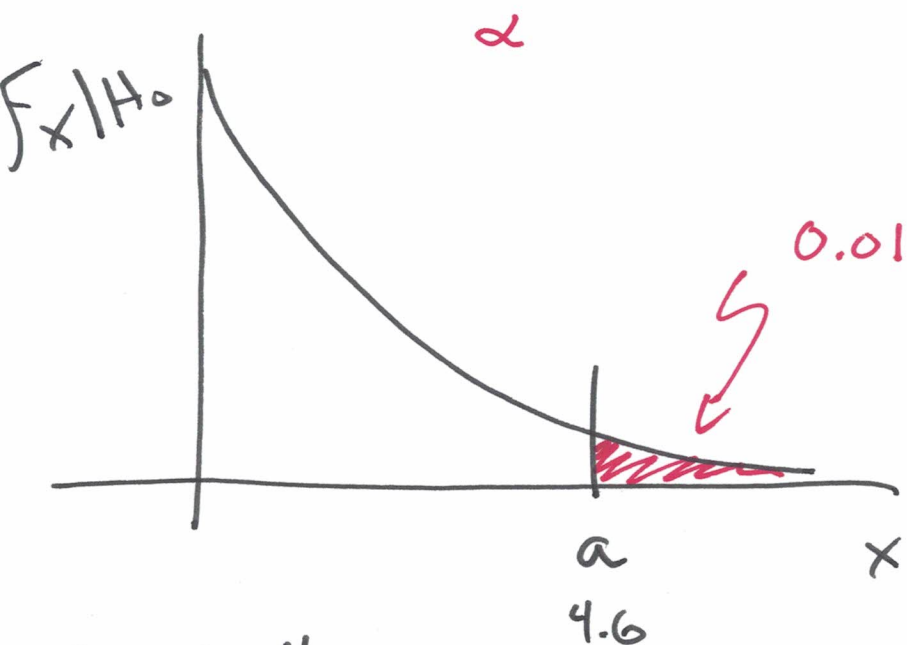
$$\alpha = \int f_X(x | H_0) dx$$

$R \leftarrow$ REJECTION REGION



EX H_0 : X IS EXPONENTIALLY DISTRIBUTED
WITH $\lambda = 1$.

- WE SEE ONE SAMPLE. DETERMINE $R = [a, \infty)$
SO THAT IF $X \in R$, REJECT H_0 AT
THE 1% SIGNIFICANCE LEVEL.



IF $X > 4.6 \Rightarrow$ REJECT H_0 .

$$f_{X|H_0}(x) = e^{-x}$$

$$F_{X|H_0} = 1 - e^{-x}$$

$$P(X > a) = e^{-a} = 0.01$$

$$-a = \ln 0.01$$

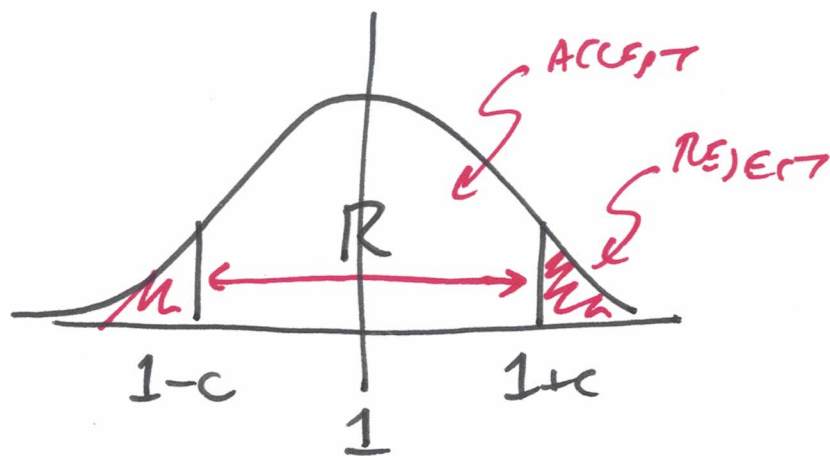
$$a = -\ln 0.01 = 4.6.$$

NOW SUPPOSE WE GET 50 I.I.D. SAMPLES
 $X_1 \dots X_N$ FROM THE UNKNOWN DISTRIBUTION.

FIND A REGION $R = [1 - c, 1 + c]$

SO THAT IF $M_{50} = \frac{1}{50} (X_1 + \dots + X_{50}) \notin R$
 THEN WE CAN REJECT H_0 AT THE 1%
 SIGNIFICANCE LEVEL.

$$E(X_i | H_0) = 1 \quad (\text{EXPONENTIAL } \lambda = 1)$$



M_{50} SHOULD LOOK
 GAUSSIAN BY THE
 C.C.T!!

CLT ROUTE:

$$P(|M_{50} - 1| > c) = 0.01$$

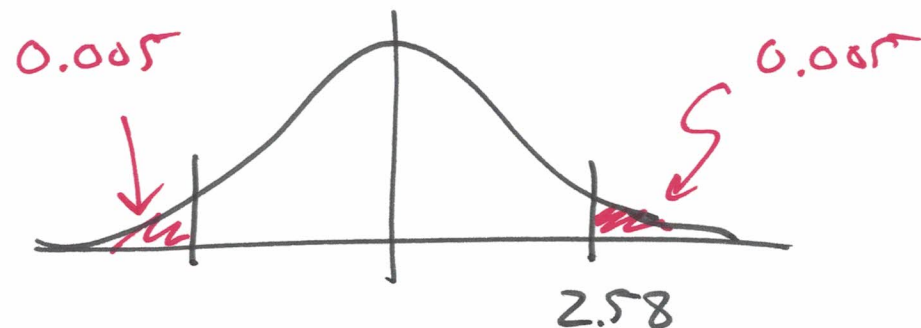
FROM CLT
EQUATION

$$\frac{Z\sigma}{\sqrt{n}} = \frac{(2.58)(1)}{\sqrt{50}}$$

$$= 0.364.$$

$\mu =$

$$[0.636, 1.364]$$



X_i IS
EXPONENTIAL
 $\lambda = 1$

$Q(2.58)$
 $= 0.005$
FROM
INVERSE
TABLE.

~~THESE~~ OUTSIDE THIS REGION, REJECT H_0 .

HYPOTHESIS TESTING:

H_0 : NULL HYPOTHESIS

$$f_X(x|H_0)$$

H_1 : ALTERNATIVE HYPOTHESIS.

$$f_X(x|H_1)$$

USUALLY H_0 = TARGET ABSENT

H_1 = TARGET PRESENT

TYPE I ERROR: DECIDE H_1 WHEN

H_0 IS ACTUALLY TRUE : FALSE ALARM

TYPE II ERROR: DECIDE H_0 WHEN

H_1 IS ACTUALLY TRUE: MISS

WE ALREADY KNOW ABOUT THE

MAXIMUM LIKELIHOOD DECISION RULE:

$$\frac{f_X(x | H_1)}{f_X(x | H_0)}$$



LIKELIHOOD

RATIO

$>$	1	$\Rightarrow$	CHOOSE H_1
$<$	1	$\Rightarrow$	CHOOSE H_0
$=$	1	$\Rightarrow$	CHOOSE EITHER.

$$\begin{matrix} H_1 \\ \geq \\ 1 \\ \leq \\ H_0 \end{matrix}$$

WE CAN DO BETTER IF WE HAVE
THE PRIOR PROBABILITIES $P(H_0)$, $P(H_1)$

WE MAY ALSO HAVE COSTS OF
MAKING THE VARIOUS DECISIONS:

C_{ij} = DECIDE j WHEN i IS TRUE.

C_{01} = FALSE ALARM COST

C_{10} = MIST COST.

USUALLY $C_{00} = C_{11} = \emptyset$, BUT MAYBE $C_{10} \neq C_{01}$
(MEDICAL TESTING, SURVEILLANCE)

BAYESIAN HYPOTHESIS TEST:

MINIMIZE EXPECTED COST $E(C)$

$$\begin{aligned} E(C) = & C_{00} P(H_0^{\text{CHOOSE}} | H_0 \text{ TRUE}) P(H_0 \text{ TRUE}) \\ & + C_{10} \dots \\ & + C_{01} \dots \\ & + C_{11} \dots \end{aligned}$$

RESULT IS THE BAYESIAN DECISION RULE:

$$\frac{f_x(x|H_1)}{f_x(x|H_0)} \begin{matrix} > \\ < \end{matrix} \begin{matrix} H_1 \\ H_0 \end{matrix} \quad \tau = \frac{P(H_0)(C_{01} - C_{00})}{P(H_1)(C_{10} - C_{11})}$$

IF $C_{00} = C_{11} = \emptyset$, $C_{01} = C_{10}$,

$$\frac{f_X(x|H_1)}{f_X(x|H_0)} \geq \frac{H_1}{H_0}$$

$$\frac{P(H_0)}{P(H_1)}$$

MAXIMUM A
POSTERIORI RULE
(MAP)

$$f_X(x|H_1)P(H_1) > f_X(x|H_0)P(H_0).$$

JUST LIKE HW 7 (WITH EXTRA COSTS).

IN FUTURE CLASSES, THINGS CAN GET MORE COMPLICATED:

H_0 : MEAN OF DISTRIBUTION IS ϕ .

H_1 : MEAN OF DISTRIBUTION IS $> \phi$

COMPOSITE HYPOTHESIS

HW 9: DOING THIS WITH SAME DATA.

FIRST STEP: ESTIMATE, $\mu, \sigma, \lambda, P(\text{BREAK}), P(H/S)$
FOR EACH DISTRIBUTION.

WHAT TO SAY IN WRITEUP:

- PROCESS FOR ESTIMATING $\mu, \sigma, \lambda \dots$
(MATH FORMULA)
- RESULT.

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

USE THIS VERSION, NOT

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2$$

$\hat{\sigma} = \sqrt{\quad}$

$\frac{1}{N-1}$ VERSION.

$\hat{\mu}$ = "ESTIMATE"

↳ WHAT'S THE DIFFERENCE?

$\frac{1}{N} \rightarrow$ ML EST OF σ^2

$\frac{1}{N-1} \rightarrow$ UNBIASED ESTIMATE

HW9 DUE LAST DAY OF CLASS APR 29

ALL DATA / COMPUTATIONAL PROBLEMS.

FINAL EXAM MAY 7 3-6 PM.